Steven Flynn

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University College London https://stevenpatrickflynn.github.io/

Research Interests

Hypoelliptic and Dispersive PDEs. Analysis on Lie groups. Microlocal Analysis. Geometric Inverse Problems. Sub-Riemannian geometry.

Employment

2024–present	**Postdoctoral Fellow**, University College London
2023–2024	**Postdoctoral Fellow**, University of Padova
2021–2023	**Postdoctoral Fellow**, University of Bath
2014–2020	**Graduate Student Instructor & Graduate Teaching Assistant**, University of California, Santa Cruz

Education

2014–2020	**Ph.D. Mathematics**, University of California, Santa Cruz (UCSC) *Dissertation title: "Unraveling Geodesic X-ray Transforms on the Heisenberg Group."* *Co-advisors:* François Monard, Richard Montgomery
2014–2015	**M.A. Mathematics**, UCSC
2012–2014	**B.A. Mathematics**, UCSC

Research Papers

Journal Articles

[1] Clotilde Fermanian-Kammerer, Véronique Fischer, and Steven Flynn. Quantization on filtered manifolds. arXiv:2412.17448, 2024. To appear in *Memoirs of the European Mathematical Society*

[2] Clotilde Fermanian-Kammerer, Véronique Fischer, and Steven Flynn. Geometric invariance of the semi-classical calculus on nilpotent graded Lie groups. *Journal of Geometric Analysis*, 2023.

[3] Steven Flynn. Injectivity of the Heisenberg X-ray transform. *Journal of Functional Analysis*, 280.5 (2021): 108886.

Refereed Proceedings and Volumes

[4] Steven Flynn. Singular Value Decomposition of the X-ray Transform on the Reduced Heisenberg group, and a Two-Radius Theorem. Springer Volume, Trends in Mathematics: Ghent Analysis and PDE Center, 2023.

[5] Clotilde Fermanian-Kammerer, Véronique Fischer, and Steven Flynn. Some Remarks on Semi-Classical Analysis on Two-Step Nilmanifolds. Springer INdAM Series, 2023.

Preprints

[6] Davide Barilari and Steven Flynn. Refined Strichartz estimates for sub-Laplacians in Heisenberg and H-type groups. arXiv:2501.04415, 2025.

[7] Steven Flynn. The sub-Riemannian X-ray Transform on H-type Groups: Fourier Slice Theorems and Injectivity Sets. arXiv:2312.00594, 2023.

Talks

Strichartz Estimates on H-type groups

May 2025	UC Santa Cruz
Jan 2025	Lund University
Dec 2024	University College London

A Microlocal Calculus on Filtered Manifolds

Dec 2025	HADES Seminar, UC Berkeley
Dec 2025	Geometric Structures Seminar, SISSA
Jan 2025	University College London
May 2024	Dispersion and Geometry in Padova

The sub-Riemannian X-ray Transform on H-type groups

Nov 2023	Junior Methusalem Seminar, Ghent University
Oct 2023	University College London
Nov 2022	Geometry and Analysis Seminar, University of Bristol

Remarks on the Semi-classical analysis of Step 2 Nilpotent groups

May 2023	Operator Algebras in the South of the UK, University of Southampton
Mar 2023	CAGE mini-seminar, Sorbonne Université

Geometric Invariance of the Semi-Classical Calculus on Graded Lie groups

Nov 2022	Conference on Noncommutative Analysis and PDEs, Queen Mary University

The Heisenberg X-ray transform

May 2022	Analysis and Differential Geometry International Seminar, University of Aveiro
Mar 2022	AGeNT Seminar, University of Bath
Nov 2021	Problèmes Spectraux en Physique Mathématique, Institut Henri Poincaré
Sep 2021	Bath Analysis Seminar, University of Bath
Apr 2021	Sub-Riemannian Seminars (online)
May 2020	UCSC Geometry and Analysis Seminar

Integral Geometry on Contact Manifolds

Jan 2020	Joint Math Meeting, Denver, Colorado
Nov 2019	Mathematical Sciences Research Institute, Berkeley, CA
Sep 2019	MSRI, Berkeley, CA

Service

Reviewer, *Journal of Functional Analysis*	2022–2023
Organizer, Bath Analysis Seminar	2022
Organizer, University of Bath Postdoc Away Day	2022
Organizer, Graduate Student Seminar, MSRI	2019
Organizer, Microlocal Analysis Seminar, UCSC	2018
Organizer, Graduate Differential Geometry Seminar, UCSC	2018

Professional Activities

Participant, MPS Workshop on LEAN	2025
Program Associate, MSRI Microlocal Analysis Program	2019
Mentor, UCSC Directed Reading Program	2018

Teaching

University College London

Autumn 2026	Analysis of the Sub-Laplacian, Taught Course Centre (Term 1, confirmed)
Autumn 2025	MATH0071: Spectral Theory

Supervision

2026–27	MSci fourth-year project, UCL
From Aug 2026	Undergraduate summer research project, UCL
2018	Mentor, Directed Reading Program, UCSC

University of California, Santa Cruz

Summer 2019	Math 105A: Real Analysis
Spring 2019	Math 105B: Real Analysis II
Summer 2018	Math 105A: Real Analysis
Summer 2017	Math 3: Precalculus

Graduate Teaching Assistant

Six years of continuous teaching experience (20 hours/week, concurrent with doctoral study). Duties included delivering lectures and review sessions, grading homework and examinations, and holding office hours.

2020 Spring	Math 105A: Real Analysis
2020 Winter	Math 105B: Real Analysis II
2019 Fall	Math 23B: Vector Calculus II
2019 Spring	Math 105B: Real Analysis II
2019 Winter	Math 105A: Real Analysis
2018 Fall	Math 152: Programming for Math
2018 Spring	Math 19B: Calculus for Science, Engineering and Mathematics II
2018 Winter	Math 100: Introduction to Proofs and Problem Solving
2017 Fall	Math 19A: Calculus for Science, Engineering and Mathematics
2017 Spring	Physics 6/6L: Introductory Physics Laboratory
2017 Winter	Math 23A: Vector Calculus
2016 Fall	Math 2: College Algebra (Collaborative Learning version)
2016 Spring	Physics 6/6L: Introductory Physics Laboratory
2016 Winter	Math 23A: Vector Calculus
2015 Fall	Math 3: Precalculus
2015 Spring	Math 11A: Calculus with Applications
2015 Winter	Math 21: Linear Algebra
2014 Fall	Math 2: College Algebra